

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member contributed ideas related to the problem of students struggling to express their academic challenges effectively.

S.No	Idea Title	Brief Description	Proposed By
1	Emotion-Aware Chatbot	A chatbot that senses frustration	Harsh
2	Facial Expression Detector	Use webcam to detect emotions	Harsh
3	Free-Text Emotion Classifier	Analyze student's written	Harsh
4	Voice Tone Analysis	Detect emotions from voice	Harsh
5	Adaptive Quiz System	Adjust quiz difficulty based on	Harsh
6	Personalized Study Planner	Generate custom study plans based on detected emotions	

and learning gaps

Harsh

Rate each idea on Impact (1-5) and Feasibility (1-5). Priority Score = Impact * Feasibility.

S.No	Idea	Impact (1-5)	Feasibility (1-5)	Priority Score	Rank
1	Emotion-Aware	4	3	12	3
2	Facial Expression	4	2	8	5
3	**Free-Text Emotion	**5**	**5**	**25**	**1**
4	Voice Tone Analysis	4	2	8	5
5	Adaptive Quiz	4	3	12	3
6	Personalized Study	5	4	20	2
	Planner				

Step 3: Selected Idea

Selected Idea: Free-Text Emotion Classifier with AI-Driven Personalized Learning Guidance

Justification:

- Highest priority score (25/25) - maximum impact with full technical feasibility
- Requires no special hardware (no camera/microphone needed)
- Directly addresses the core problem: students can describe their study difficulties in plain text
- Enables real-time, personalized AI guidance using Google Gemini API
- Supports 5 emotion classes: Bored, Confident, Confused, Curious, Frustrated
- Can be deployed as a free Streamlit web application accessible from any device

Step 4: Refined Idea Statement

> "An AI-powered platform that accepts a student's free-text description of a study problem, detects their emotional state using a dual BiLSTM + BERT ensemble model enhanced by rule-based keyword matching, and generates personalized learning guidance using the Google Gemini API - all presented through a dark-themed, interactive Streamlit web application."